Compiled Solutions – Intermediate Division

1. **00-01 C4 Prefix/Infix/Postfix**

   The equal symbol is treated as any other operator. As shown
   $AX^2 + BX + C = 0$
   A * (^X 2) + (* B X) + C = 0
   (* A ^ X 2) + (* B X) + C = 0
   + + * A ^ X 2 * B X C = 0
   = + + * A ^ X 2 * B X C 0

2. **00-01 C4 Prefix/Infix/Postfix**

   The expression translates as follows: 3
   $[\ (|2 - 6|)^{\frac{1}{2}} + ((3 + 4)^2)^{1/2}\ ]^{1/2} = (9)^{1/2} = 3$

3. **01-02 C4 Prefix/Infix/Postfix**

   The given expression converts to infix as follows: As shown
   A((A + B) / (C – (A / B)))
   This converts to prefix as follows: * A / + A B - C / A B

4. **01-02 C4 Prefix/Infix/Postfix**

   Converting the formula to infix produces the following: As shown
   V = 4 / 3 * π * R ↑ 3
   This converts to prefix as follows: = V * * / 4 3 π ↑ R 3

5. **02-03 C4 Prefix/Infix/Postfix**

   / / * 2 + 4 6 5 - ↑ 3 2 5 = / / * 2 (4 + 6) 5 - (↑ 3 2) 5 = 1
   / / (2 * 10) 5 (9 – 5) = / (20 / 5) (4) = 4 / 4 = 1

6. **02-03 C4 Prefix/Infix/Postfix**

   $(A + B)^{2/3} + (A / 2 - B) / 3 =$ As shown
   A B + 2 3 / ↑ A 2 / B – 3 / +

7. **03-04 C4 Prefix/Infix/Postfix**

   / ↑ + A B 2 * C + ↑ A 2 B translates to $\frac{(A+B)^2}{C(A^2+B)}$ As shown
   which translates to A B + 2 ↑ C A 2 ↑ B + * /

8. **03-04 C4 Prefix/Infix/Postfix**

3 2 ↑ 2 3 * - 2 2 ↑ @ 8 2 – 6 3 / * @ = 13
(3^2) (2 * 3) – (2^2) @ (8- 2)(6/3) * @ =
$\sqrt{3^2+4^2}$ @ 12 = $\sqrt{5^2+12^2}$ = 13

9. **04-05 C4 Prefix/Infix/Postfix**

8 4 / 6 2 ↑ + 2 + 2 3 ↑ / = (8/4) (6↑2) + 2 + (2↑3) / = 5
2 36 + 2 + 8 / = (2 + 36) 2 + 8 / = 38 2 + 8 / =
(38 + 2) 8 / = 40 / 8 = 5

10. **04-05 C4 Prefix/Infix/Postfix**

+ / - ↑ A B 2 - C A * A B = + / - (↑ A B) 2 (- C A) (* A B) = As shown
+ / - (A↑ B) 2 (C- A) (A* B) = + / (A↑ B – 2) (C- A) (A* B) =
(A↑ B – 2) / (C - A) + (A* B).
The last set of parentheses is not required and should not be included.

11. **05-06 C4 Prefix/Infix/Postfix**

$AB + AB - *A2 \uparrow B2 \uparrow - =$ translates to infix as As shown
$(A+B)*(A-B) = A \uparrow 2 - B \uparrow 2$ and this translates to prefix as
$= * + AB - AB - \uparrow A2 \uparrow B2$

12. **05-06 C4 Prefix/Infix/Postfix**

$/**@\,AB@\,AC@\,BC*@\,AD@\,BD$ translates to 96

$$\frac{(A@B)*(A@C)*(B@C)}{(A@D)*(B@D)} = \frac{(12@40)(12@24)(40@24)}{(12@2)(40@2)} =$$

$$\frac{4*12*8}{2*2} = 96$$

13. **06-07 C4 Prefix/Infix/Postfix**

* / - / A B C + / A B C + B C converts to the following As shown
postfix expression: A B / C – A B / C + / B C + *

14. **06-07 C4 Prefix/Infix/Postfix**

The expression translates to: 1

$$\frac{(A^2)^2-(B^2)^2}{(A^2-B^2)(A^2+B^2)} = \frac{3^4-2^4}{(3^2-2^2)(3^2+2^2)} = \frac{65}{65} = 1$$

15. **07-08 C4 Prefix/Infix/Postfix**

A B * C A /+ A B ↑ - B C / + converts to the following prefix expression: + - + * A B / C A ↑ A B / B C — As shown

16. **07-08 C4 Prefix/Infix/Postfix**

+ - + 3 / 4 2 / 8 + 3 1 * 6 2 = + - + 3 (4 / 2) / 8 (3 + 1) (6*2) = + - (3 + 2) (8 / 4) 12 = + (5 – 2) 12 = 3 + 12 = 15 — 15

17. **08-09 C4 Prefix/Infix/Postfix**

- + - ↑ X 3 * 3 ↑ X 2 * 3 X 2 = - + - (↑ X 3) * 3 (↑ X 2) (* 3 X) 2 — 0
= - + - X^3 (* 3 X^2) (3*X) 2 = - + (- X^3 (3 * X^2)) (3 * X) 2
= - + ((X^3 - 3*X^2) (3*X)) 2 = - (X^3 – 3 * X^2 + 3 * X) 2
= X^3 – 3 * X^2 + 3 * X – 2
X = 2 ⇒ 2^3 – 3 * 2^2 + 3 * 2 – 2 = 0

18. **08-09 C4 Prefix/Infix/Postfix**

4 2 + 3 / 3 2 8 + * 5 / @ 2 2 ↑ @ = (4 2 +) 3 / 3 (2 8 +) * 5 / @ 2 2 ↑ @ — 6
= (6 3 /)(3 10 *) 5 / @ 2 2 ↑ @ = 2 (30 5 /) @ (2 2 ↑) @
= (2 6 @) 4 @ = 6 4 @ = 6

19. **09-10 C4 Prefix/Infix/Postfix**

$\frac{A+B^2}{C} - C*(B+\frac{A}{C}) \Rightarrow \frac{AB2\uparrow+}{C} - C*(BAC/+) \Rightarrow$ — As shown
(AB2 ↑ +C/) – (CBAC/ + *) ⇒ AB2 ↑ +C/CBAC/ + * –

20. **09-10 C4 Prefix/Infix/Postfix**

-/ + # 12 20 * 6 4 - ↑ 2 3 1 / # 150 25 ↑ 5 2 — 3
= - / + (12 # 20) (6 * 4) – (2^3) 1 / (150 # 25) (5^2)
= - / (4 + 24) (8 – 1) (25/25) = 28 / 7 - 1 = 4 – 1 = 3

21. **10-11 C4 Prefix/Infix/Postfix**

$\frac{(A+B)C}{A*B^2-\frac{B}{C}}$ = ((A + B) * C) / (A * B^2 – B / C) — As shown
= (A B + C *) / (A B ↑ 2 * B C / -)
= A B + C * A B 2 ↑ * B C / - /

22. **10-11 C4 Prefix/Infix/Postfix**

* + # 9 2 # 8 3 ↑ 3 2 4 = # * + (# 9 2) (# 8 3) (↑3 2) 4 — 13
= # * (+ 4 2) 9 4 = # (* 6 9) 4 = # 54 4 = 13

23. **11-12 C2 Prefix/Infix/Postfix**

4 + 5 * 8 – 6 / 2 – 7 = 4 + (5 * 8) – (6 / 2) – 7 — As shown
= (4 + (5 8 *)) – (6 2 /) – 7 = ((4 5 8 * +) – (6 2 /)) – 7
= (4 5 8 * + 6 2 / –) – 7 = 4 5 8 * + 6 2 / – 7 –

24. **11-12 C2 Prefix/Infix/Postfix**

+ * 4 ↑– / 6 2 1 2 * / 8 2 + – ↑ 3 2 8 1 — 24
= + * 4 ↑– (/ 6 2) 1 2 * (/ 8 2) + – (↑ 3 2) 8 1
= + * 4 ↑ (– 3 1) 2 * 4 + (– 9 8) 1 = + * 4 (↑ 2 2) * 4 (+ 1 1)
= + (* 4 4) (* 4 2) = + 16 8 = 24

25. **12-13 C2 Prefix/Infix/Postfix**

A^2 + B * (A – B)^2 – C / A * (B^2 – C) — As shown
= (↑ A 2) + B*(↑ − A B 2) – (/ C A) * (− ↑ B 2 C)
= (↑ A 2) + (* B ↑ − A B 2) – (* / C A − ↑ B 2 C)
= (+↑ A 2 * B ↑ −A B 2) – (* / C A − ↑ B 2 C)
= − +↑ A 2 * B ↑ − A B 2 * / C A −↑ B 2 C

26. **12-13 C2 Prefix/Infix/Postfix**

2 3 9 + * 6 / 8 2 / 6 4 + * 5 / - = 2 (3 9 +) * 6 / (8 2 /) (6 4 +) * 5 / − — -4
= (2 12 *) 6 / (4 10 *) 5 / −
= (24 6 /) (40 5 /) –
= 4 8 − = -4

27. **13-14 C2 Prefix/Infix/Postfix**

$\frac{A^2(B+C)}{A-C}+\frac{A+B}{C^2}$ = (((A^2) * (B+C)) / (A – C)) + ((A + B) / C^2) — As shown
= (((A 2 ↑) * (B C +)) / (A C -)) + ((A B +) / (C 2 ↑))
= ((A 2 ↑ B C + *) / (A C -)) + (A B + C 2 ↑ /)
= (A 2 ↑ B C + * A C - +) + (A B + C 2 ↑)
= A 2 ↑ B C + * A C - / A B + C 2 ↑ / +

28. **13-14 C2 Prefix/Infix/Postfix**

+ + / * 3 ↑ + 2 8 2 ↑ 5 2 4 / * 6 8 * 4 3 — 20
= + + / * 3 ↑ (+ 2 8) 2 (↑ 5 2) 4 / (* 6 8) (* 4 3)
= + + / * 3 (↑ 10 2) 25 4 (/ 48 12)
= + + / (* 3 100) 25 4 4
= + + (/ 300 25) 4 4
= + (+ 12 4) 4 = + 16 4 = 20

29. **14-15 C2 Prefix/Infix/Postfix**

$A^2 \frac{2B}{C} + \frac{A}{B+C} - \frac{A}{B}$ — As shown

= (↑ A 2) + ((* 2 B) / C) + (A / (+ B C)) – (/ A B)
= ((↑ A 2) + (/ * 2 B C))+ (/ A + B C) – (/ A B)
= ((+ ↑ A 2 / * 2 B C) + (/ A + B C)) – (/ A B))
= ((+ + ↑ A 2 / * 2 B C / A + B C) – (/ A B))
= − + + ↑ A 2 / * 2 B C /A + B C / A B

30. **14-15 C2 Prefix/Infix/Postfix**

bdiff 4 bdiff 3 1 8 6 = bdiff 4 (bdiff 3 1 8) 6 — 3
= bdiff 4 (max(|3-1|, |1-8|, |3-8|) 6
= bdiff 4 7 6
= max (|4-7|, |7-6|, |4-6|)
= 3

31. **15-16 C2 Prefix/Infix/Postfix**

$(s * (s - a) * (s - b) * (s - c))^{1/2}$ — As shown
= ((s * (– s a)) * (– s b) * (– s c)) ↑ (/ 1 2)
= (((* s – s a) * (− s b)) * (− s c)) ↑ (/ 1 2)
= ((* * s – s a – s b) * (− s c)) ↑ (/ 1 2)
= (* * * s – s a – s b – s c) ↑ (/ 1 2)
= ↑ * * * s – s a – s b – s c / 1 2

32. **15-16 C2 Prefix/Infix/Postfix**

2 3 2 ↑ − 4 8 2 / 1 + * 2 / + 2 3 * 6 / + — 4
= 2 (3 2 ↑) – 4 (8 2 /) 1 + * 2 / + (2 3 *) 6 / +
= (2 9 –) 4 (4 1 +) * 2 / + (6 6 /) +
= -7 (4 5 *) 2 / + 1 +
= -7 (20 2 /) + 1 +
= (-7 10 +) 1 +
= 3 1 + = 4